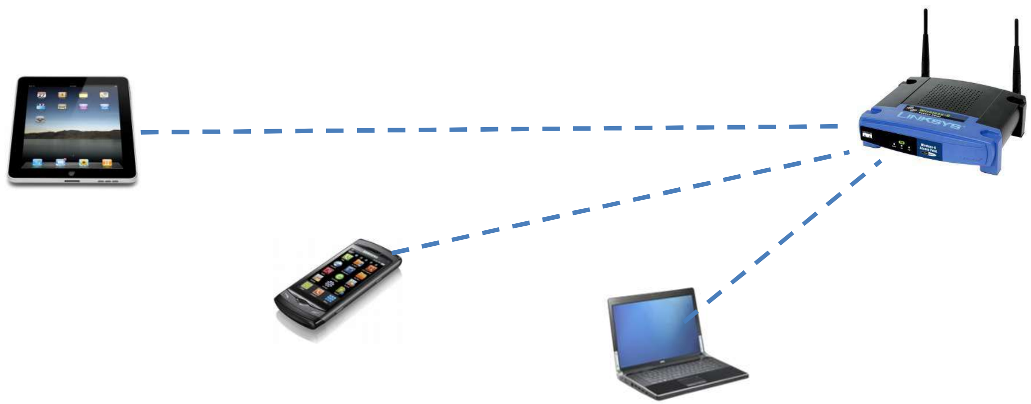
Case Study: Wi-Fi

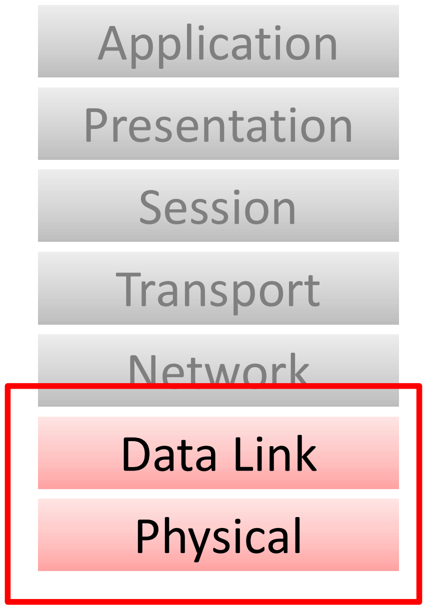
What is Wi-Fi?

- A name for local area wireless computer networking technology
- Based on IEEE 802.11 standards on Wireless Local Area Networks (WLANs)
- Used as a synonym for WLAN (Wireless LANs)

IEEE 802.11 standards

- Defines physical and data link layer techniques
- Physical layer
  - Modulation, coding techniques
- Data link layer
  - Medium access control





802.11 standards: The main stream

- Legacy 802.11 (1997)
  - 2.4GHz band, 2 Mbps
  - Frequency hopping spread spectrum (FHSS)
    - Patented by Hollywood star Hedy Lamarr

- 802.11b (1999)
  - 2.4GHz band, 11Mbps
  - Direct sequence spread spectrum (DSSS)
- 802.11a (1999)
  - 5GHz band, 54Mbps
  - Orthogonal frequency division multiplexing (OFDM)

802.11 standards: The main stream

- 802.11g (2003)
  - 2.4GHz band, but OFDM and 54Mbps
  - compatible with 802.11b

- 802.11n (2009)
  - 2.4GHz and 5GHz band
  - OFDM
  - up to 600Mbps with 40MHz channel, 4x4 MIMO

- 802.11ac (2014)
  - Gigabit throughput
  - Wider RF bandwidth, more MIMO streams, high-density modulation

802.11 standards: The main stream

- 802.11ax (Wi-Fi 6, 2019)
  - 2.4GHz / 5GHz / 6GHz (Wi-Fi 6E) bands
  - OFDMA: multiple users transmit simultaneously on different subcarriers
  - MU-MIMO (uplink and downlink)
  - 1024-QAM
  - Up to ~9.6 Gbps
- 802.11be (Wi-Fi 7, 2024)
  - 4096-QAM, 320 MHz channel
  - Multi-Link Operation (MLO): use multiple bands simultaneously
  - Up to ~46 Gbps
- 802.11bn (Wi-Fi 8, expected 2028)
  - Focus on Ultra High Reliability (UHR), not just speed
  - Multi-AP coordination
  - Targeting emerging applications (AR/VR, industrial IoT)

802.11 standards: others

- 802.11e: QoS enhancements
- 802.11i: security
- 802.11k: radio resource management
- 802.11p: vehicular environments (WAVE)
- 802.11s: mesh networking
- 802.11u: interworking with cellular systems
- 802.11ad/ay: 60 GHz, multi-Gbps, short range (e.g., VR, wireless docking)
- 802.11ah (Wi-Fi HaLow): sub-1GHz, long-range, low-power IoT
- 802.11bb (Li-Fi): light-based communication
- 802.11bf: WLAN Sensing - using Wi-Fi signals to sense the environment (motion, presence, breathing, etc.)

- ... many others

Wireless Channel

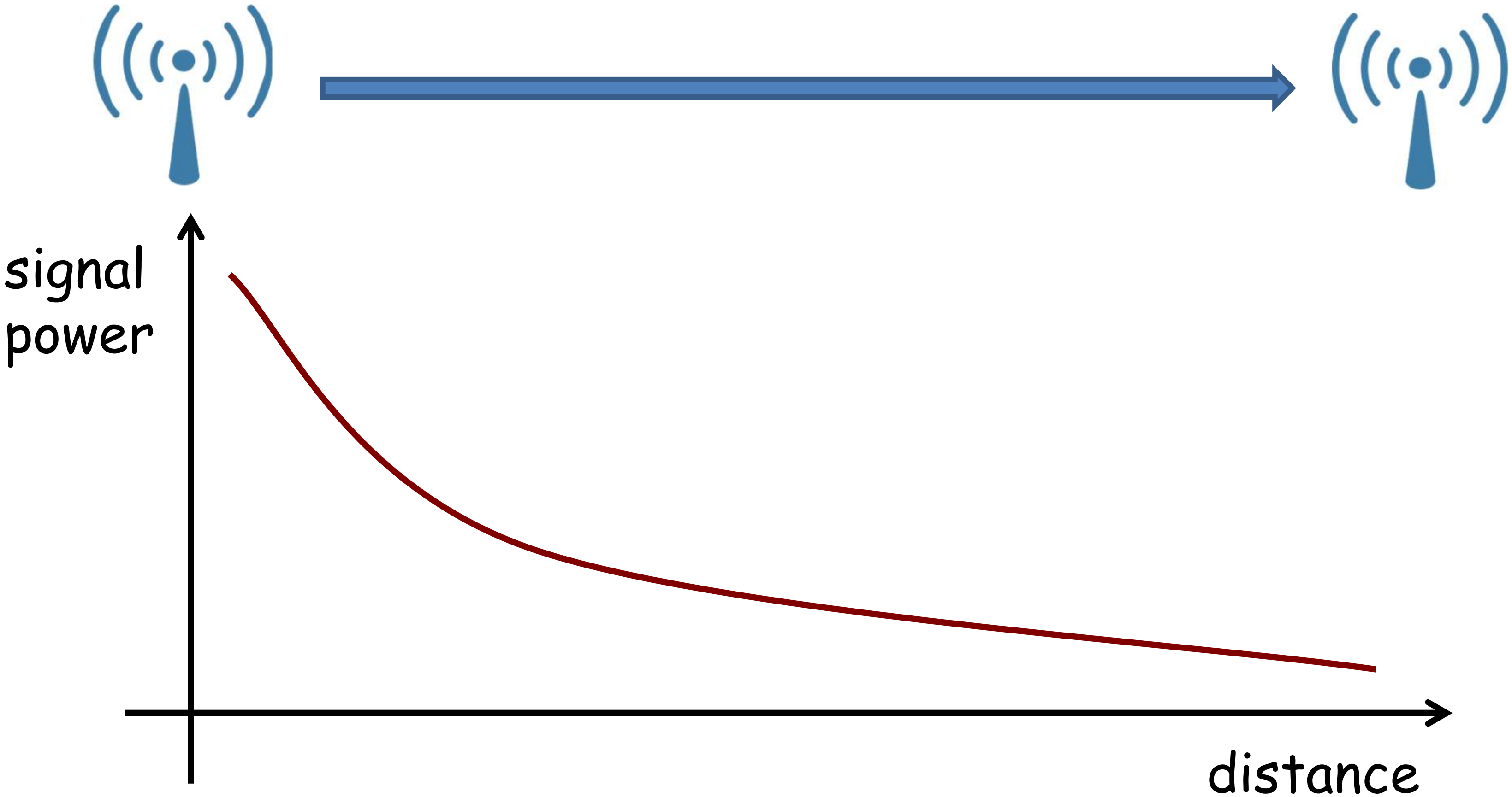
Wireless Channel

- Signal transmitted onto the wireless channel from a transmit (TX) antenna

- Signal travels through the air and is received at the receive (RX) antenna

Signal Attenuation

- As the signal travels through the air, its power is attenuated



Pathloss

- The received power (P_r) is a function of transmit power (P_t) and distance (R)
- Friis propagation model (free space):

$$\frac{P_r}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2$$

- General log-distance pathloss model

$$P_r \propto \frac{1}{R^n}$$

  - n: pathloss exponent - depends on the environment
    - Free space: $n = 2$
    - Urban: $n = 3 \sim 4$
    - Indoor (with obstacles): $n = 4 \sim 6$
- Signal attenuation based on distance is called pathloss
- G_t = transmit antenna gain, G_r = receive antenna gain, λ = wavelength

SNR and Decoding

- Background noise (noise floor): present at the receiver due to hardware and environment

- SNR (Signal-to-Noise Ratio): ratio of signal power to noise power

$$\mathrm{SNR} = \frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}$$

  – sometimes SINR (Signal-to-Interference-plus-Noise Ratio) is used

$$\mathrm{SNR} = \frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}} + P_{\mathrm{interference}}}$$

- For successful decoding, SNR must exceed a minimum required SNR

- Required SNR depends on the modulation and coding (MCS)
  – Lower MCS (e.g., BPSK 1/2): low required SNR
  – Higher MCS (e.g., 64-QAM 5/6): high required SNR

Receiver Sensitivity

- Receiver sensitivity: the minimum signal power required for successful decoding
- Determined by:
  - $\text{Sensitivity (dBm)} = \text{Noise Floor (dBm)} + \text{Required SNR (dB)}$
- Since required SNR depends on MCS, sensitivity also varies by MCS

10.2 WLAN 2.4 GHz Receiver Performance Specifications

Note: Unless otherwise specified, the specifications in Table 17 are measured at the chip port (for the location of the chip port, see Figure 16 on page 36).

Table 17. WLAN 2.4 GHz Receiver Performance Specifications

Parameter	Condition/Notes	Minimum	Typical	Maximum	Unit
Frequency range	–	2400	–	2500	MHz
RX sensitivity (8% PER for 1024 octet PSDU) [a]	1 Mbps DSSS	–97.5	–99.5	–	dBm
	2 Mbps DSSS	–93.5	–95.5	–	dBm
	5.5 Mbps DSSS	–91.5	–93.5	–	dBm
	11 Mbps DSSS	–88.5	–90.5	–	dBm
RX sensitivity (10% PER for 1000 octet PSDU) at WLAN RF port [a]	6 Mbps OFDM	–91.5	–93.5	–	dBm
	9 Mbps OFDM	–90.5	–92.5	–	dBm
	12 Mbps OFDM	–87.5	–89.5	–	dBm
	18 Mbps OFDM	–85.5	–87.5	–	dBm
	24 Mbps OFDM	–82.5	–84.5	–	dBm
	36 Mbps OFDM	–80.5	–82.5	–	dBm
	48 Mbps OFDM	–76.5	–78.5	–	dBm
	54 Mbps OFDM	–75.5	–77.5	–	dBm
	20 MHz channel spacing for all MCS rates (Mixed mode)				

Fading

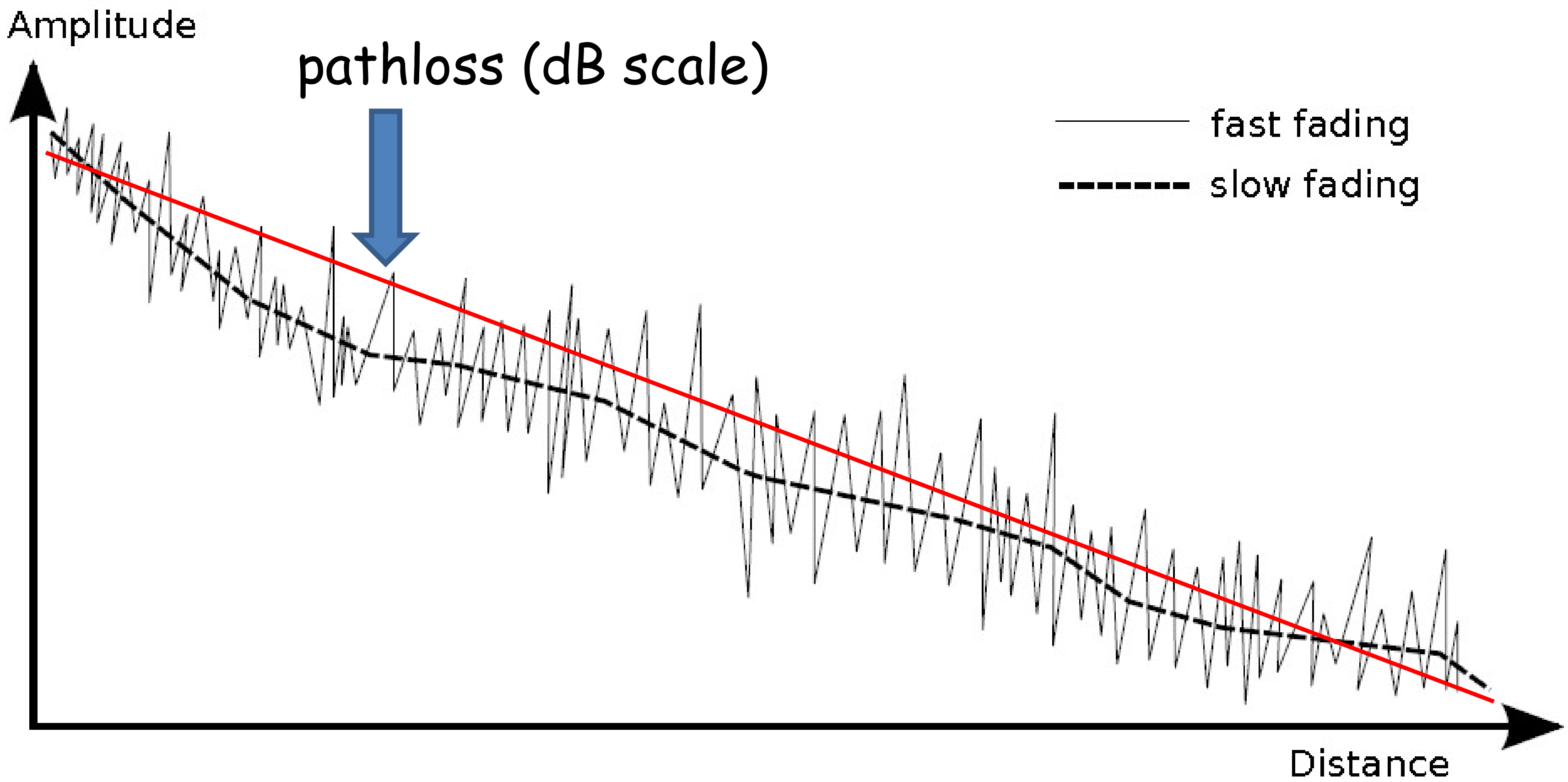
- Factors affecting received signal power
  - Pathloss
  - Fading
- Fading
  - Deviation of attenuation affecting a signal
- Types of fading
  - Slow fading (large-scale fading, shadowing)
  - Fast fading (small-scale fading)

Slow vs. Fast Fading

- Slow (large-scale) fading
  - Cased by large obstruction (hill, large building, wall)
  - Also called shadowing
  - Time scale: seconds or longer (e.g., walking behind a building)
  - Modeled using log-normal distribution

- Fast (small-scale) fading
  - Caused by multipath propagation
    - Signal reflects off walls, floors, objects
    - Multiple copies arrive with different delays and phases
  - Time scale: milliseconds or less (changes with small movements)
  - Amplitude and phase fluctuate rapidly over the period of use

Pathloss and Fading



Wi-Fi: Physical Layer Aspects

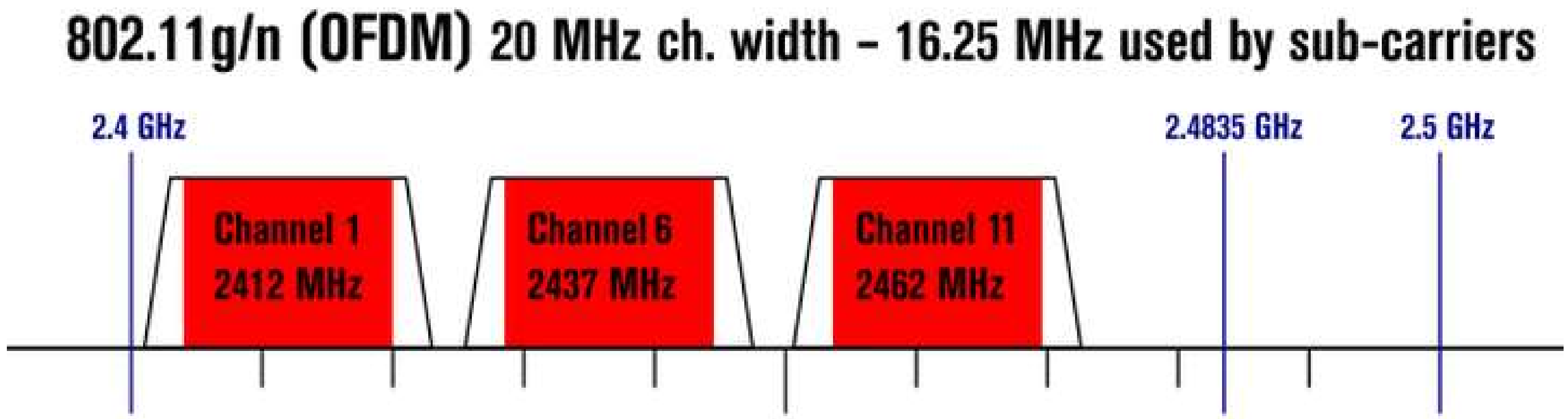
Physical Layer

- Signal generation from bits
- Recovery of bits from the received signal
- Tightly coupled with physical medium
- Modulation and coding scheme (MCS)

Modulation

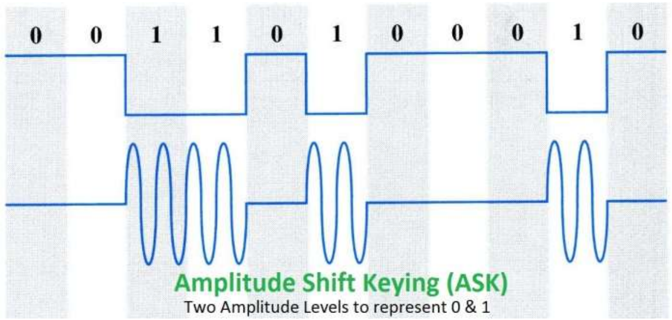
- Process of varying one or more properties of carrier signal with a modulating signal which contains information

- Carrier signal: a high frequency signal for carrying data
  - E.g. WLAN channels



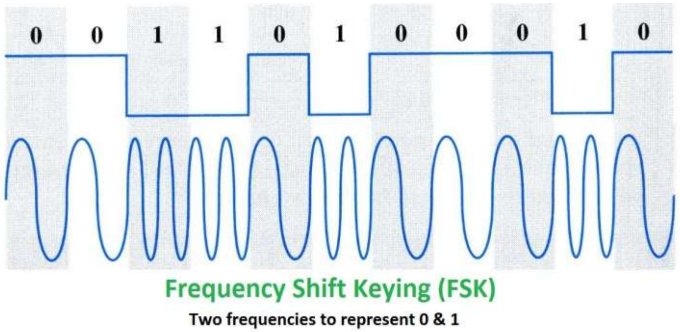
Modulation

- ASK

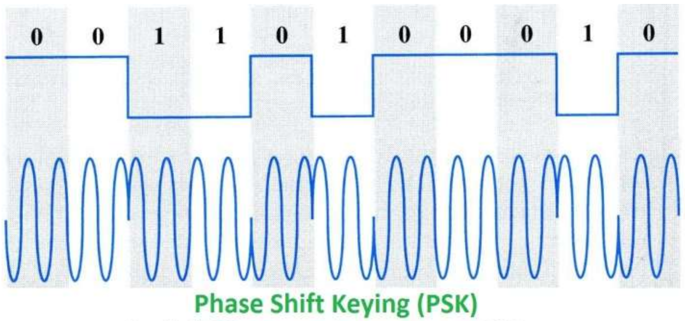


symbol

- FSK

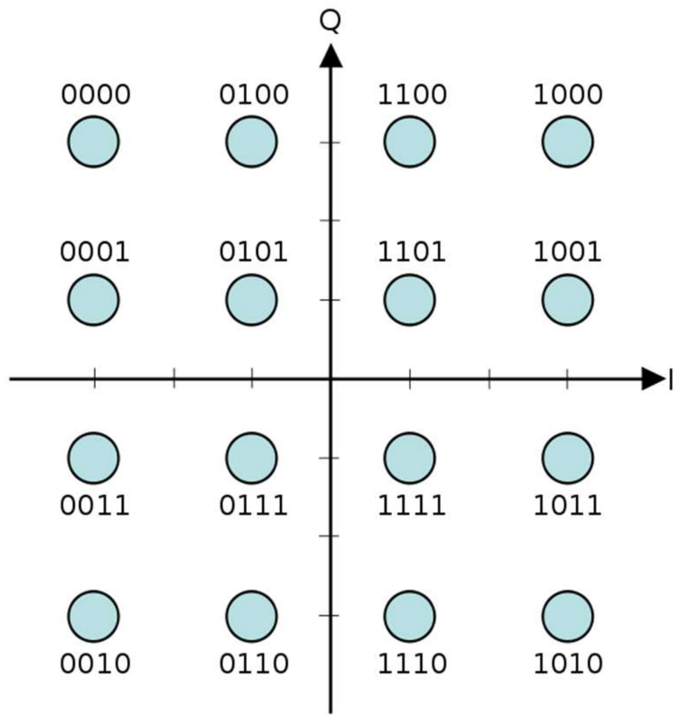


- PSK



Modulation

- QAM (Quadrature Amplitude Modulation)
  - PSK + ASK



constellation diagram for 16-QAM

Modulation used in Wi-Fi

- BPSK: 1 bit per symbol (2 signals)
- QPSK: 2 bits per symbol (4 signals)
- 16-QAM: 4 bits per symbol (16 signals)
- 64-QAM: 6 bits per symbol (64 signals)
- 256-QAM: 8 bits per symbol (256 signals) — used in 802.11ac
- 1024-QAM: 10 bits per symbol (1024 signals) — used in Wi-Fi 6 (802.11ax)
- 4096-QAM: 12 bits per symbol (4096 signals) — used in Wi-Fi 7 (802.11be)

Coding (Channel Coding)

- Insert error-correcting codes (ECC) in order to make the message tolerable to errors

- E.g. Hamming (7,4) code
  - Corrects 1-bit error in a 7-bit message

- Coding rate
  - Ratio of data bits in the message
  - E.g. 3/4 coding rate: 75% data, 25% ECC

- Codes used in Wi-Fi
  - Convolutional codes (802.11a/g/n/ac)
  - LDPC (Low-Density Parity-Check) codes (Wi-Fi 6 and later)

MCS (Modulation and Coding Scheme)

- Defines modulation scheme and coding rate
  - The transmitter should tell the receiver which MCS level it is using

802.11n MCS Table

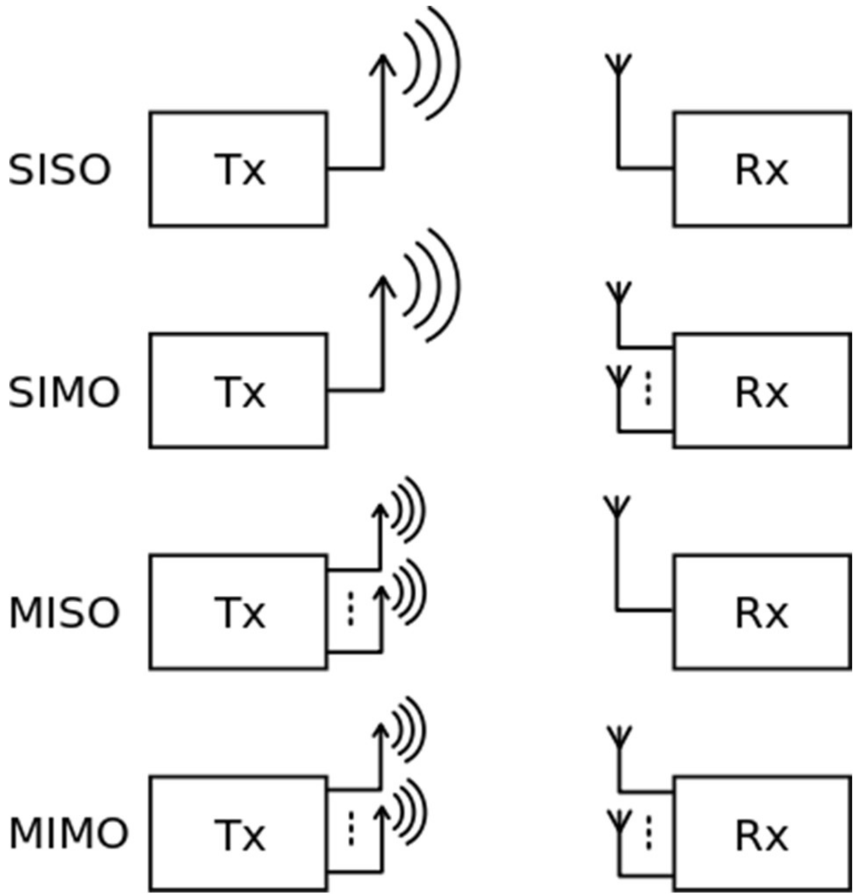
MCS Index	Type	Coding Rate	Spatial Streams	Data Rate (Mbps) with 20 MHz CH		Data Rate (Mbps) with 40 MHz CH	
				800 ns	400 ns (SGI)	800 ns	400 ns (SGI)
0	BPSK	1 / 2	1	6.50	7.20	13.50	15.00
1	QPSK	1 / 2	1	13.00	14.40	27.00	30.00
2	QPSK	3 / 4	1	19.50	21.70	40.50	45.00
3	16-QAM	1 / 2	1	26.00	28.90	54.00	60.00
4	16-QAM	3 / 4	1	39.00	43.30	81.00	90.00
5	64-QAM	2 / 3	1	52.00	57.80	108.00	120.00
6	64-QAM	3 / 4	1	58.50	65.00	121.50	135.00
7	64-QAM	5 / 6	1	65.00	72.20	135.00	150.00
8	BPSK	1 / 2	2	13.00	14.40	27.00	30.00
9	QPSK	1 / 2	2	26.00	28.90	54.00	60.00
10	QPSK	3 / 4	2	39.00	43.30	81.00	90.00
11	16-QAM	1 / 2	2	52.00	57.80	108.00	120.00
12	16-QAM	3 / 4	2	78.00	86.70	162.00	180.00
13	64-QAM	2 / 3	2	104.00	115.60	216.00	240.00
14	64-QAM	3 / 4	2	117.00	130.00	243.00	270.00
15	64-QAM	5 / 6	2	130.00	144.40	270.00	300.00
16	BPSK	1 / 2	3	19.50	21.70	40.50	45.00
...	...	...	...	...	...	...	...
31	64-QAM	5 / 6	4	260.00	288.90	540.00	600.00

MCS (Modulation and Coding Scheme)

- Selection of MCS level depends on the link quality
- Higher MCS level requires higher SNR
- Since the SNR can change dynamically over time, MCS level is changed adaptively
- This adaptive process is called rate adaptation (or auto-rate selection)

MIMO (Multiple Input Multiple Output)

- Use of multiple antennas at both the transmitter and the receiver to improve communication performance



MIMO: advantages

- Spatial multiplexing gain: improves spectral efficiency (data rate)
  - How: transmit different data streams on each antenna (in parallel)
  - Receiver separates the streams using channel knowledge

- Diversity gain: improves link reliability
  - How: transmit the same data through multiple independent paths
  - Achieved by space-time coding (e.g., Alamouti) — no precise channel info needed at TX

- Array gain (beamforming): improves SNR/range
  - How: transmit the same signal with carefully adjusted phases on each antenna
  - Signals add constructively at the receiver
  - Requires channel state information (CSI) for phase calculation

MIMO: Spatial Multiplexing Gain

- M transmit antennas, N receive antennas

- Spatial multiplexing gain: $\min(M, N)$
  - Number of parallel data streams that can be transmitted

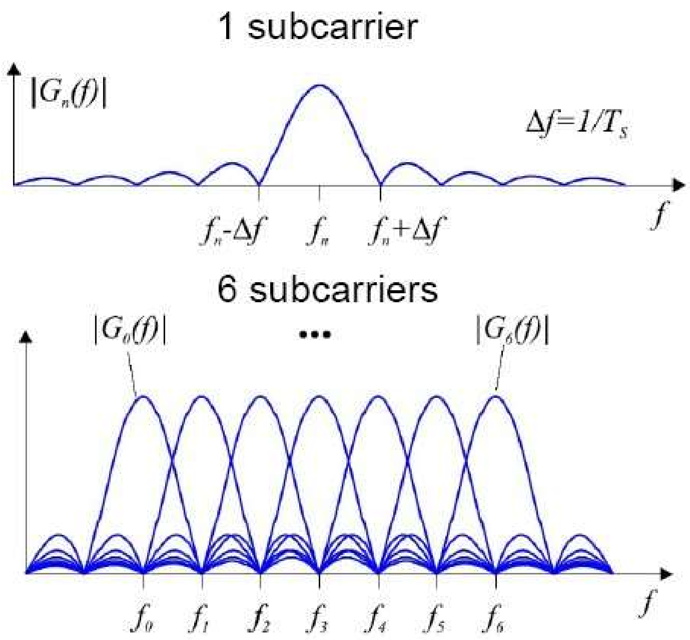
- E.g., 2 TX antennas and 4 RX antennas
  - $\min(2, 4) = 2$
  - 2 parallel data streams → up to 2x data rate compared to SISO

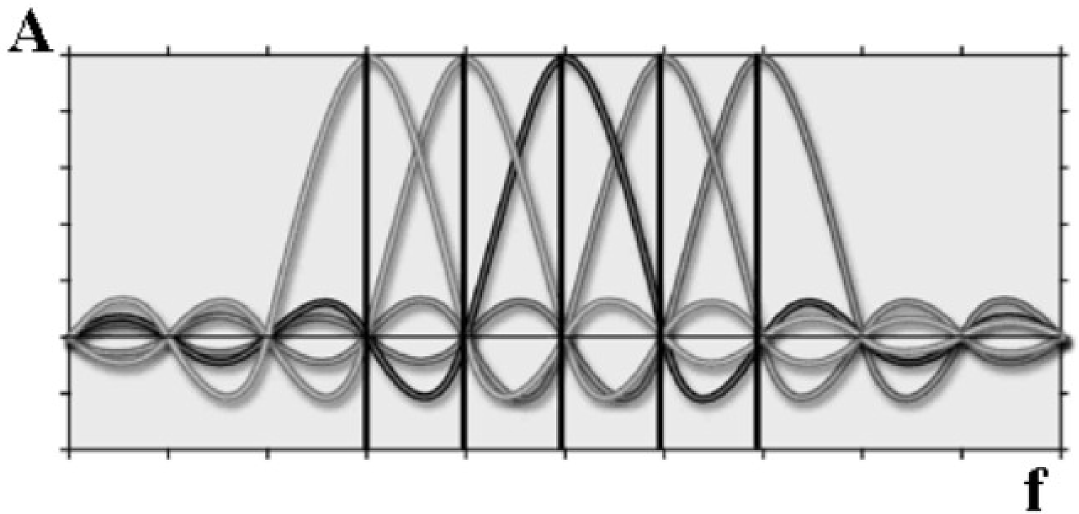
- The smaller of (M, N) limits the multiplexing gain
  - 2x4 system can only transmit 2 streams (limited by TX side)
  - 4x4 system can transmit 4 streams

Physical layer: OFDM

- A specialized FDM (frequency division multiplexing)

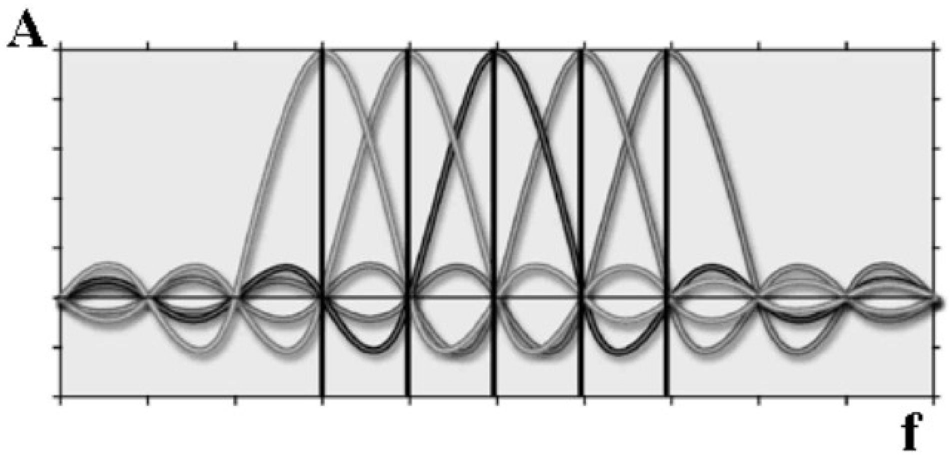
- Sub-carriers are chosen so that they are orthogonal to each other – they do not interfere with each other





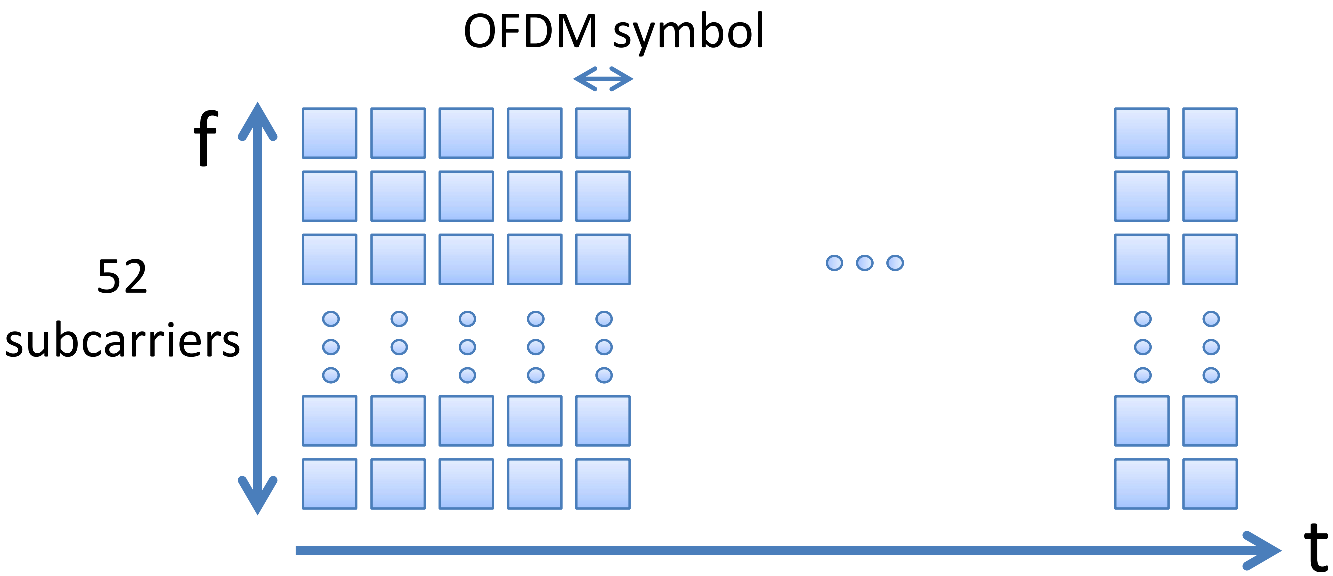
Why OFDM?

- Can easily adapt to channel conditions
- Robust against narrow-band co-channel interference
- Robust against inter-symbol interference (ISI) and fading caused by multipath propagation
- High spectrum efficiency compared to spread spectrum techniques



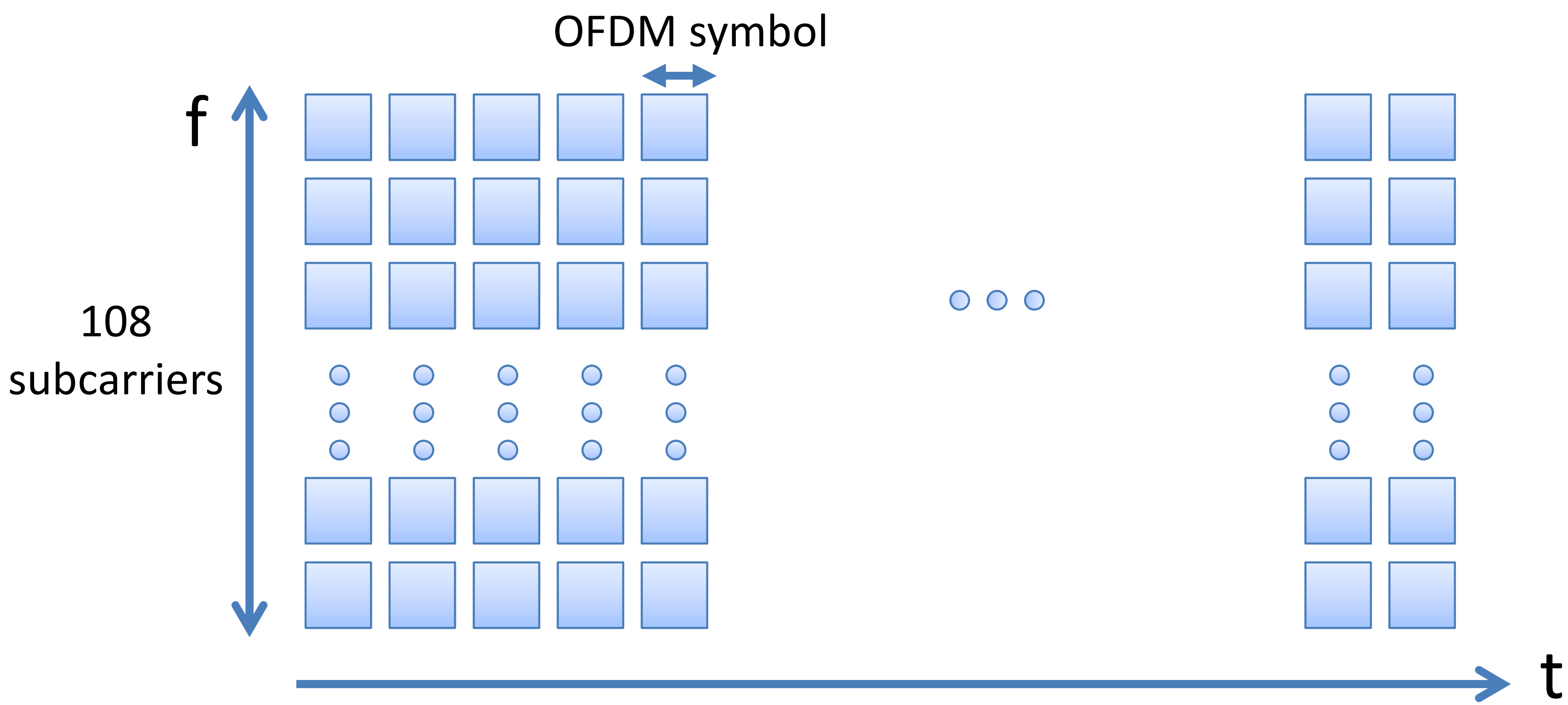
802.11n OFDM – 20MHz channel

- 20MHz channel, divided into 56 subcarriers
  - 52 subcarriers used for data, 4 subcarriers for pilot
  - Each subcarrier: 312.5kHz, signal bandwidth: 17.5MHz
- Symbol structure
  - Data portion: 3.2us (=1/312.5kHz, satisfying orthogonality condition)
  - Guard interval (cyclic prefix): 0.8us (Long Gi), 0.4us (Short GI)
- Guard Interval: protects against inter-symbol interference from multipath propagation



802.11n OFDM – 40MHz channel

- 40MHz channel, divided into 114 subcarriers
  - 108 subcarriers used for data, 6 subcarriers for pilot
  - Each subcarrier: 312.5kHz, signal bandwidth: 35.625MHz
  - Symbol duration: 3.2us + guard interval (0.8us or 0.4us)



802.11n OFDM – Modulation & Coding

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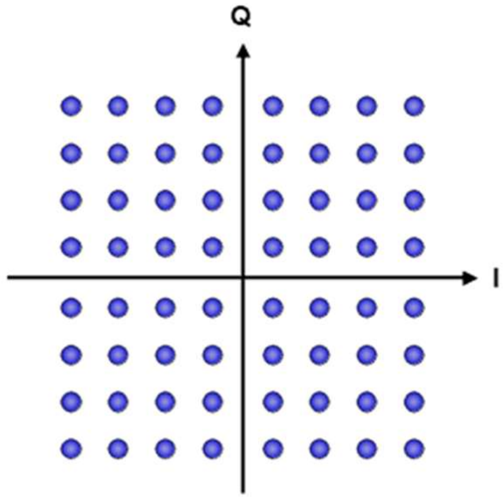
...	...	...	...	...	...	...	...
31	64-QAM	5 / 6	4	260.00	288.90	540.00	600.00

Achieving 600Mbps

- Modulation: 64QAM
  - 6 bits are coded per subcarrier
- Coding: 5/6
  - 16.6% redundant bits for error correction
- Data bits per OFDM symbol (40MHz)
  - 6 x (5/6) x 108 subcarriers = 540 bits
- Using Short Guard Interval (3.6us symbol)
  - Number of OFDM symbols per second = 277,777
- Data rate: 540 x 277,777 = 150Mbps

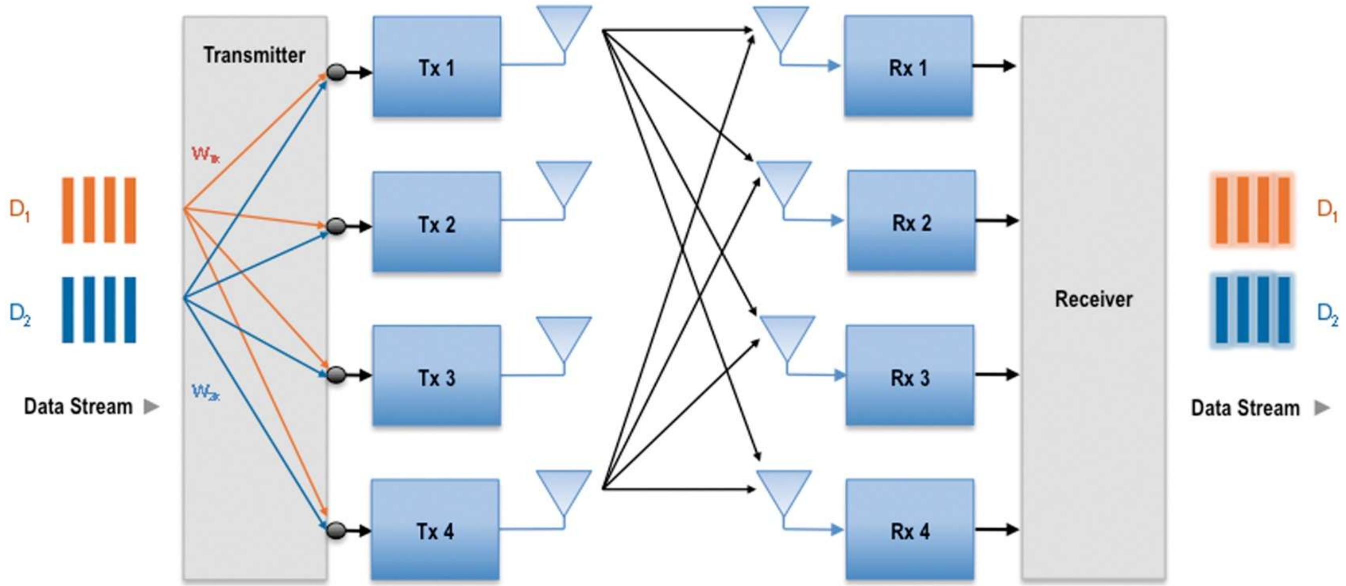


64 QAM

Achieving 600Mbps

- Spatial multiplexing: send different streams on multiple antennas
  - Multiplexing gain = $\min(M, N)$

- 4x4 MIMO → 4 parallel streams → 4x capacity compared to SISO
- Use 4x4 MIMO: 150Mbps → 600Mbps



Evolution of Wi-Fi Data Rates

	802.11n (Wi-Fi 4)	802.11ac (Wi-Fi 5)	802.11ax (Wi-Fi 6)	802.11be (Wi-Fi 7)
Year	2009	2014	2019	2024
Max channel width	40MHz	160MHz	160MHz	320MHz
Highest modulation	64-QAM	256-QAM	1024-QAM	4096-QAM
Max spatial streams	4	8	8	16
Max data rate	600Mbps	~6.9Gbps	~9.6Gbps	~46Gps

- Three dimensions of improvement
  - Wider channels: more frequency bandwidth
  - Higher modulation: more bits per symbol
  - More spatial streams: more parallel data flows

End of Class